BHUMIKA DEVENDRA SURYAWANSHI

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SUMMARY

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP and Generative AI. Experienced in developing LLM/RAG applications and AI solutions using Python, TensorFlow, Scikit-learn, OpenCV, MediaPipe, LangChain, LangGraph and FastAPI.

EDUCATION

Post Graduate Certificate Program in Artificial Intelligence — **Feb 2026 - Aug 2026**
CDAC Mumbai
Percentage: 79%

Bachelor of Engineering in Computer Science (SPPU) — **2021-2025**
CGPA: 7.84

HSC (Arts, Commerce, Science Junior College, Sinnar) — **2020-2021**
Percentage: 90%

SSC (Loknete Shankarrao Balaji Waje Vidyalaya, Sinnar) — **2018-2019**
Percentage: 85%

PROJECTS

TOWARDS RESPONSIBLE AI: QUANTIFYING COGNITIVE LOAD IN HUMAN-AI COLLABORATION

Tools & Technologies:
Python, Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV, MediaPipe, LibreFace, Flask, Machine Learning, Deep Learning, Computer Vision, Multimodal AI, Feature Engineering, Responsible AI
Developed a multimodal AI system for cognitive-load classification using computer vision, eye tracking, mouse tracking and emotion recognition, with engineered behavioral features to classify cognitive load into Low, Medium and High levels.
Implemented real-time behavioral feature extraction, machine learning classification and Responsible AI principles to support adaptive AI-assisted learning.

ABNORMAL ACTIVITY DETECTION USING AI

Tools & Technologies:
Python, OpenCV, MediaPipe Pose, Scikit-learn, K-Nearest Neighbors (KNN), Flask, Firebase, NumPy, Pandas, Multithreading, Computer Vision, Machine Learning, Pose Estimation, Human Activity Recognition, Real-Time Processing, Text-to-Speech
Developed a real-time Computer Vision system for human activity recognition and abnormal posture detection using MediaPipe Pose, OpenCV and KNN classification.
Integrated Flask and Firebase with multithreaded processing to deliver real-time voice and email alerts.

WEATHER PREDICTION USING MACHINE LEARNING

Tools & Technologies:
Python, Scikit-learn, Pandas, NumPy, XGBoost, Random Forest, FastAPI, Machine Learning, Data Preprocessing, Feature Engineering, Model Evaluation, Model Deployment, REST API
Built a Machine Learning-based weather prediction system using Random Forest and XGBoost, achieving 96%+ accuracy after data preprocessing and feature engineering.
Deployed the trained ML model using FastAPI REST APIs for real-time predictions.

TECHNICAL SKILLS

- Programming Languages: Python, SQL, Core Java
- Machine Learning & AI: Supervised Learning, Classification, Regression, Feature Engineering, Data Preprocessing, Scikit-learn, KNN, Logistic Regression, Decision Trees, Random Forest, XGBoost
- Deep Learning: ANN, CNN, TensorFlow, PyTorch, Model Training, Model Evaluation
- Generative AI & NLP: Generative AI, LLMs, NLP, RAG, Prompt Engineering, LangChain, LangGraph, AI Agents, Embeddings, Vector Databases, Semantic Search
- Computer Vision: Computer Vision, OpenCV, MediaPipe, Pose Estimation, Image Processing, Facial Landmark Detection, Emotion Recognition, Human Activity Recognition, Real-Time Video Processing
- Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Cleaning, Data Preprocessing, Data Visualization
- Backend & Deployment: Flask, FastAPI, REST API, Model Deployment, Firebase
- Database: MySQL, Firebase, DBMS, PostgreSQL(PgVector)
- Tools / DevOps: Git, GitHub, Docker

CERTIFICATIONS

- Core Java Training(2025) from FUEL Institute -Basics of Java and OOP
- Aptitude Training(2025) from FUEL Institute - Quantitative Aptitute, Logical reasoning
- NPTEL Big Data Computing(2024): 67%
- Digital Image Processing(2024) from Great Learning
- PreCAT-Preparatory Course for Entrance Exam(2026) by Sunbeam

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LANGUAGES

- **English**: Professional working proficiency
- **Hindi**: Native/Full professional proficiency
- **Marathi**: Native/Full professional proficiency